



**iRA
Synergy**

— Private Limited —

Nashik, Maharashtra, India

DUAL-PURPOSE (2-IN-1) DEAD ANIMAL & SOLID WASTE INCINERATOR

BUILT FOR A CLEANER TOMORROW

**CAPACITY:
500 KG TO 1 TON PER DAY**



ECO
FRIENDLY



HIGH
EFFICIENCY



LOW
EMISSION



AUTOMATIC
OPERATION



COST
EFFECTIVE



CLEAN ENVIRONMENT



SAFE DISPOSAL



SUSTAINABLE FUTURE

ABOUT iRA SYNERGY

iRA Synergy Private Limited is committed to providing advanced, reliable and environmentally responsible waste management solutions. Our 2-in-1 Incinerator is engineered for safe disposal of dead animals and solid waste with advanced emission control and high efficiency.



Advanced Technology



Robust Construction



Low Operating Cost



Environmental Compliance



MULTI-ANGLE PRODUCT VIEW



FRONT VIEW



REAR VIEW



LEFT VIEW



RIGHT VIEW



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HEAVY DUTY
CONSTRUCTION



EASY
MAINTENANCE



LOW FUEL
CONSUMPTION



AUTOMATIC
OPERATION



LONG
SERVICE LIFE

BUILT FOR A
CLEANER TOMORROW



HOW IT WORKS – ADVANCED 5-CHAMBER TECHNOLOGY

- 1 PRIMARY COMBUSTION CHAMBER**
High temperature combustion begins
- 2 SECONDARY COMBUSTION CHAMBER**
Complete destruction of smoke, odor & pathogens
- 3 HEAT EXCHANGER CHAMBER**
Reduces flue gas temperature
- 4 WET SCRUBBER CHAMBER**
Gas cleaning and pollution control
- 5 ID FAN & CHIMNEY STACK**
Cleaned gas released to atmosphere



COMPLETE
COMBUSTION



HIGH DESTRUCTION
EFFICIENCY



SMOKE & ODOR
CONTROL



CLEAN & SAFE
EMISSION



ENVIRONMENT
FRIENDLY

KEY FEATURES & ADVANTAGES



Dual Purpose: Dead Animal & Solid Waste Disposal



Heavy Duty Steel Construction



High Temperature Combustion (800°C – 1100°C)



Low Fuel Consumption



Advanced Emission & Smoke Control System



Ash Collection System



Diesel / Gas Burner Option



Easy Maintenance



PLC Based Automatic Operation



Long Service Life



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BUILT FOR A
CLEANER TOMORROW



ECO
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HIGH
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AUTOMATIC
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COST
EFFECTIVE

TECHNICAL SPECIFICATIONS

PARAMETER	DETAILS
 Model	AWI-500
 Capacity	500 kg/batch (900 kg – 1 Ton per day)
 Combustion Chamber	4.5 m (L)
 Secondary Chamber	2.5 m (L)
 Chimney Height	30.0 meters
 Temperature	800°C – 1100°C
 Fuel Type	Diesel / Gas / LPG / PNG
 Power Supply	380V, 3 Phase, 50Hz
 Control System	PLC Automatic
 Emission Standard	As per CPCB Norms



High Temperature Combustion



Low Fuel Consumption



Advanced Smoke Control



Robust & Durable Construction



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HIGH EFFICIENCY



LOW EMISSION



AUTOMATIC OPERATION



COST EFFECTIVE



BUILT TO LAST

IDEAL FOR THE DISPOSAL OF

Safe, Hygienic & Environmentally Responsible Waste Management Solution



Dead Animals



Poultry Waste



Meat & Bone Waste



Veterinary Waste



Municipal Waste



Animal By-products
& Offal



Hatchery Waste



Slaughterhouse Waste

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SAFE & HYGIENIC
Ensures complete
destruction of waste
& pathogens



**ENVIRONMENT
FRIENDLY**
Reduces pollution
& protects nature



HIGH EFFICIENCY
Advanced 5-chamber
technology for
maximum efficiency



COST EFFECTIVE
Low operating cost
with high
performance



COMPLIANT
Meets CPCB norms
& environmental
standards

ENVIRONMENT BENEFITS



Complete combustion ensures zero harmful emissions



Reduces air pollution & greenhouse gas emissions



Helps in maintaining clean & healthy environment



Ash output is minimal, sterile & safe to dispose



Supports sustainable waste management practices



CERTIFICATIONS & COMPLIANCES



CPCB
Type Approval
Certificate



MPCB
Compliance



9001:2015
Quality Management



14001:2015
Environment
Management



CE
Certification



NABL
Accredited
Laboratory



SWM
Solid Waste
Management Rules
Compliance



WILDLIFE
Wildlife Disposal
Rules Compatibility

GST

GST Registration
Certificate

PAN

PAN Registration
Certificate

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AUTHENTIC
CERTIFICATIONS



REGULATORY
COMPLIANCE



ENVIRONMENT
FRIENDLY

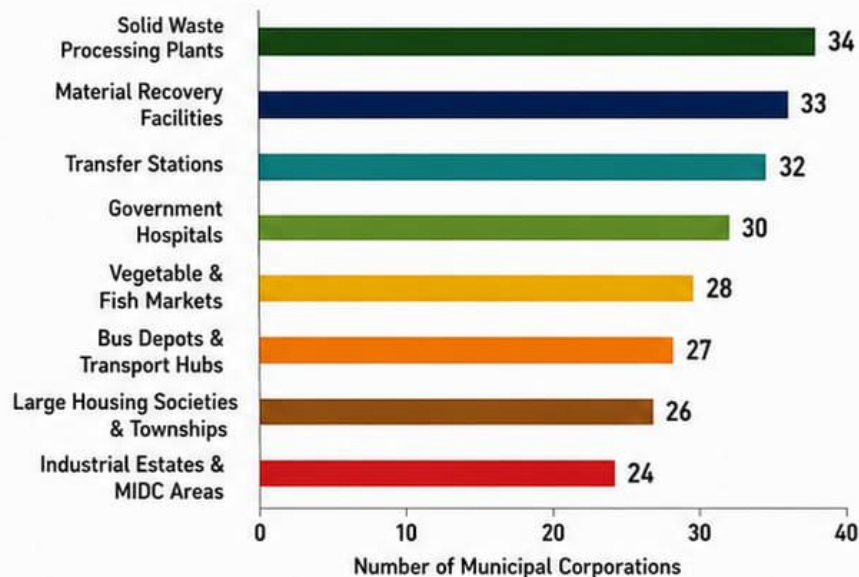


TRUSTED
PERFORMANCE

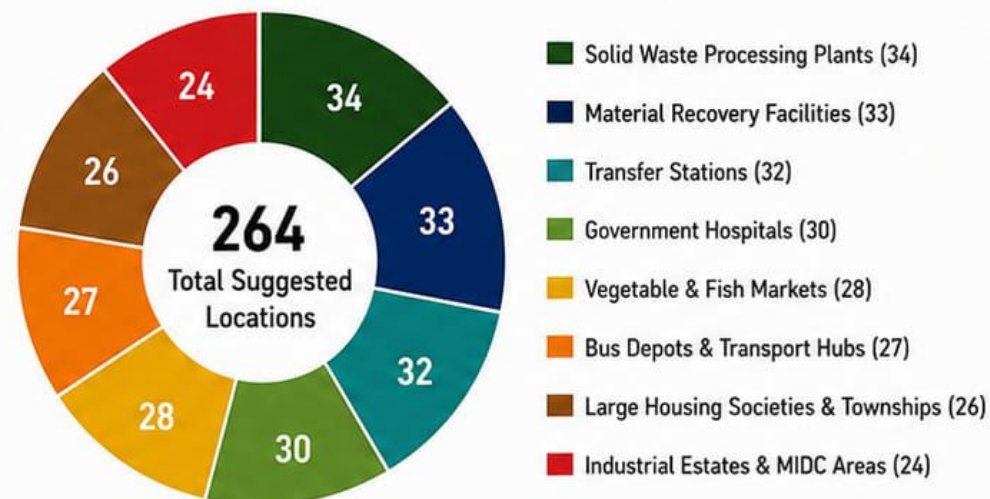
SUGGESTED LOCATIONS & PLACES IN UD DEPARTMENT – ALL MUNICIPAL CORPORATIONS OF MAHARASHTRA



SUGGESTED LOCATIONS BY CATEGORY



SUGGESTED LOCATIONS – DISTRIBUTION



FOCUS AREAS FOR UD DEPARTMENT



APPLICATION AREAS



Municipal
Corporations



Veterinary
Hospitals



Poultry
Farms



Dairy
Farms



Slaughterhouses



Livestock
Institutions



Government
Institutions



Animal
Husbandry Dept.



Veterinary
Colleges



Waste Management
Organizations

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WIDE RANGE
OF APPLICATIONS



EFFECTIVE WASTE
MANAGEMENT



SAFE & HYGIENIC
SOLUTION



ENVIRONMENTALLY
RESPONSIBLE



TRUSTED ACROSS
INDUSTRIES

—• BUILT TO MEET GOVERNMENT STANDARDS —•

- ✓ Complies with CPCB Guidelines for Animal Waste Incineration
- ✓ Aligned with Solid Waste Management Rules, 2016
- ✓ Supports Clean India & Swachh Bharat Abhiyan
- ✓ Eligible for State & Central Government Approvals & Subsidies*

*As per applicable policies



CONSTRUCTION & SAFETY



Heavy Duty MS Steel Construction



High-grade Refractory Lining



Automatic Control Panel with
Temperature Monitoring



Safety Interlocks & Alarms



Easy Operation & Maintenance





PROCESS FLOW DIAGRAM

SMOKE CONTROL SYSTEM



1

WASTE INPUT

Waste is collected and fed into the system.

2

PRIMARY & SECONDARY COMBUSTION CHAMBER

Waste is burned in a controlled environment to reduce volume and destroy harmful substances.

3

HEAT EXCHANGER CHAMBER

Hot flue gases pass through the heat exchanger to transfer heat and cool down.

4

WET SCRUBBER CHAMBER

Gases are scrubbed with water or chemical solution to remove pollutants and particulates.

5

ID FAN

Induced Draft (ID) fan pulls the cleaned gas through the system and ensures proper airflow.

6

CHIMNEY STACK

Treated and cleaned gases are released into the atmosphere through the chimney stack.

SYSTEM BENEFITS



Reduces Air Pollution



Ensures Environmental Compliance



Waste Volume Reduction



High Operational Efficiency



Sustainable & Eco-Friendly Solution

WHY CHOOSE iRA SYNERGY 2-IN-1?



High Efficiency
Performance



Advanced Emission
Control



Robust & Durable
Construction



Cost-Effective
Operation



Environmentally
Responsible



Easy Maintenance
& Long Life



Automatic
& Reliable Operation



Low Operating
Cost



ARCHITECTURAL STRUCTURE LAYOUT DESIGN – INCINERATOR UNIT

DEAD ANIMAL & SOLID WASTE INCINERATOR PLANT

PROJECT COMPONENTS

1. INCINERATOR UNIT

- Incinerator Capacity : 400 kg to 1 Ton / hr
- High temperature primary chamber for complete combustion.
- Secondary chamber for gas burnout.
- Suitable for dead animals & solid waste disposal.
- Robust design with refractory lining and insulation.

2. STACK ASSEMBLY

- CPCB compliant Stack with Scrubber, Dust Collector and 30 Meter Stack Height.
- Continuous emission monitoring provision.

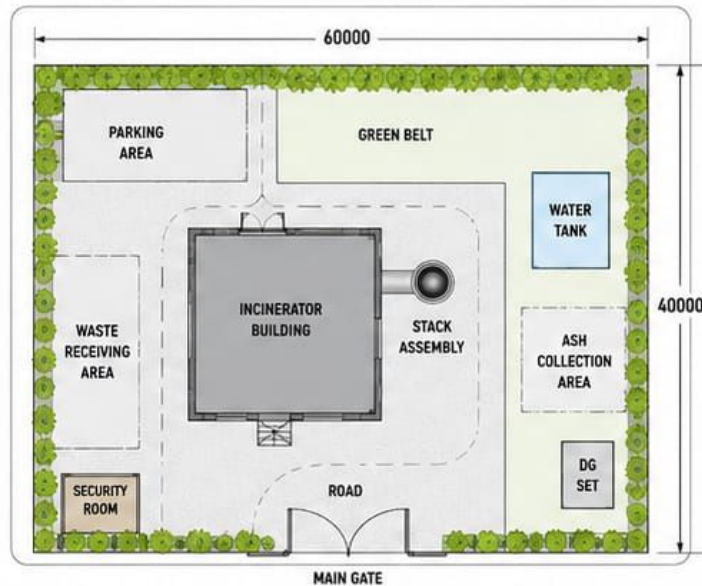
CIVIL WORKS

- RCC Foundation as per structural design.
- Drainage system for wastewater collection.
- Storm water drain and soak pit arrangement.
- Paved area with interlocking / RCC road for plant operation.

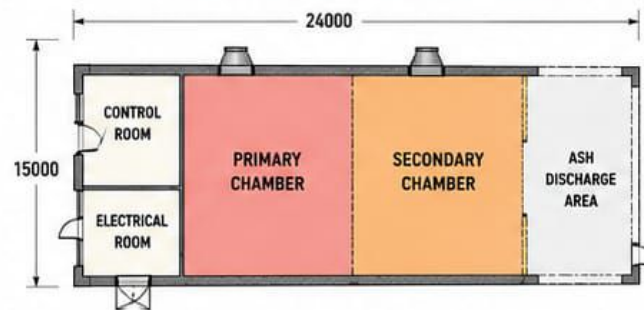
ELECTRIFICATION & INSTALLATION

- HT/LT electrical panel, control panel and earthing system.
- Cable trenching and wiring for all equipment.
- PLC based control system with automation.
- Complete installation, testing and commissioning.

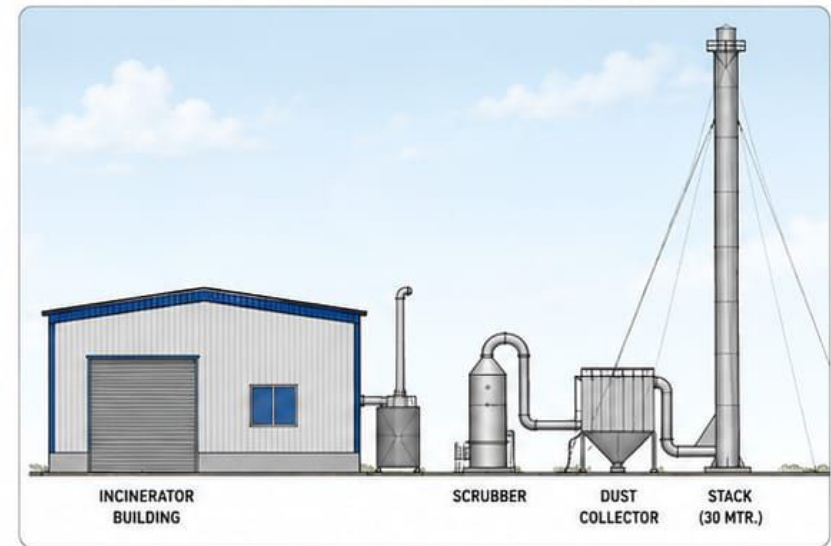
SITE LAYOUT PLAN



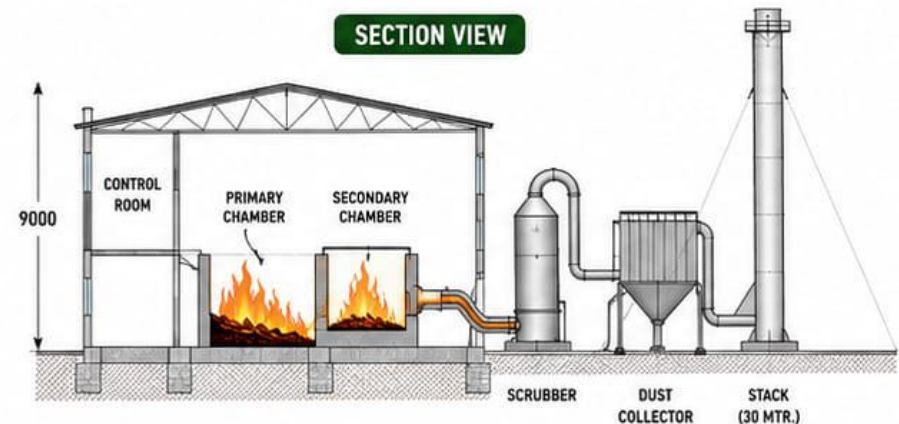
FLOOR PLAN (INCINERATOR BUILDING)



FRONT ELEVATION



SECTION VIEW



SAFE & HYGIENIC
Environment friendly disposal solution



POLLUTION CONTROL
CPCB compliant system with emission control



AUTOMATED OPERATION
PLC based system for efficient operation



DURABLE & RELIABLE
Long life, low maintenance and robust structure



SUITABLE FOR
Municipal Corporations, ULBs & Panchayats

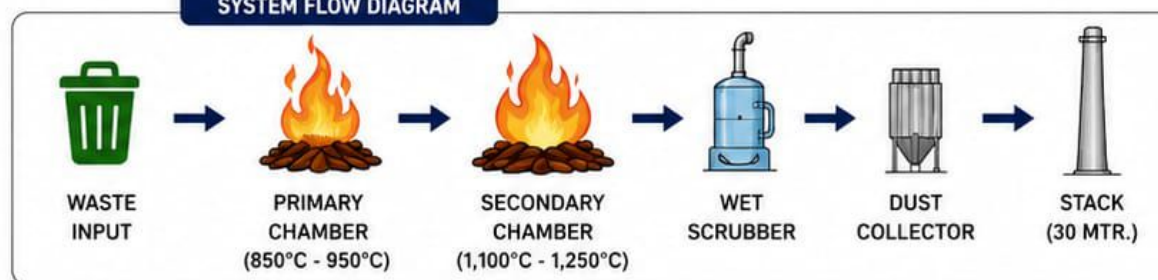
iRA Synergy Private Limited

TECHNICAL DATASHEET – DUAL CHAMBER PYROLYTIC INCINERATOR

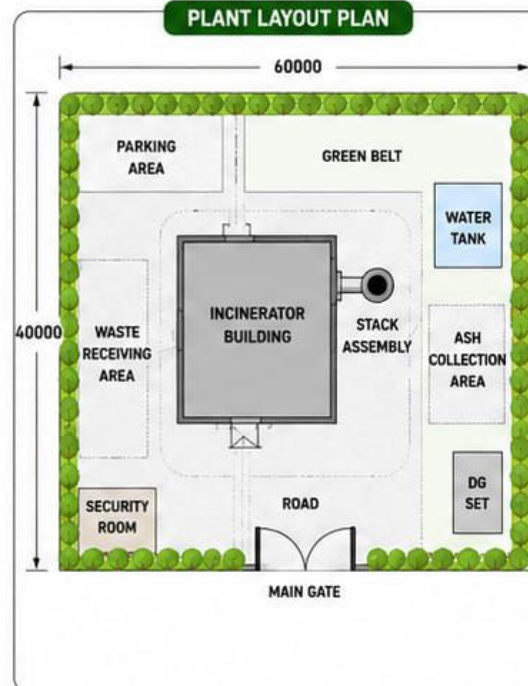
TECHNICAL PARAMETERS & STANDARD SPECIFICATIONS

SR. NO.	PARAMETER	SPECIFICATION / DETAIL	ICON
1	Type	Dual Chamber Pyrolytic Incinerator System	
2	Capacity	400 kg to 2,000 kg per day (2 Tons/day)	
3	Primary Chamber Temperature	850°C to 950°C (Optimal for Complete Combustion)	
4	Secondary Chamber Temperature	1,100°C to 1,250°C (For Gas & Smoke Combustion)	
5	Retention Time	Minimum 2 Seconds (in Secondary Chamber)	
6	Fuel Options	Diesel / LPG / PNG (Other: As per Availability)	
7	Construction Material	6–10 mm Thick Mild Steel (MS) and High Grade Refractory Brick Lining (Up to 1,400°C)	
8	Air Pollution Control	Wet Scrubber, Cyclone Dust Collector and Stack System	
9	Stack/Chimney Height	30 Meter High Stack (CPCB Compliant)	
10	Control Panel	PLC Based with Digital HMI Display and IOT Based Data Restoring	
11	Residues	5% to 10% of Total Waste (Ash Output)	
12	Pollution Control Compliance	CPCB & MPCB Compliant System (Meets 100% Emission Norms)	

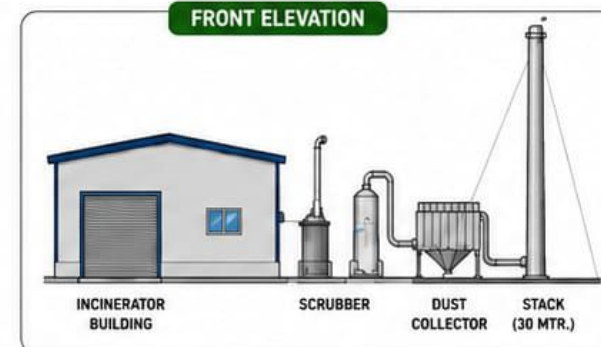
SYSTEM FLOW DIAGRAM



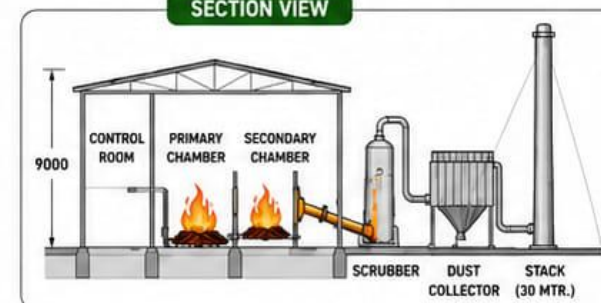
PLANT LAYOUT PLAN



FRONT ELEVATION



SECTION VIEW



HIGH TEMPERATURE COMBUSTION

Ensures complete waste destruction with high temperature efficiency



LOW EMISSION

CPCB & MPCB compliant system for clean environment



AUTOMATED OPERATION

PLC based system with efficient for easy and safe disposal



SAFE & HYGIENIC

Environment friendly solution for bio-medical, municipal & animal waste



iRA
Synergy
— Private Limited —



MAKE IN INDIA

SUSTAINABLE SOLUTIONS FOR A **CLEANER TOMORROW**

Advanced Dual Chamber Pyrolytic Incinerator
System for safe, efficient and eco-friendly
waste management solutions.



ECO FRIENDLY
SOLUTIONS



ADVANCED
TECHNOLOGY



POLLUTION
CONTROL



SUSTAINABLE
FUTURE



HIGH
EFFICIENCY

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Nashik, Maharashtra, India



**HIGH TEMPERATURE
COMBUSTION**
Ensures complete
waste destruction



LOW EMISSION
CPCB & MPCB
compliant system



**AUTOMATED
OPERATION**
PLC based system
with IoT monitoring



MINIMUM RESIDUE
Only 5% to 10%
ash output



SAFE & HYGIENIC
Environment friendly
waste management

**INNOVATING TODAY
FOR A BETTER TOMORROW**